

# Pradeep Yenkuwale

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## PROFESSIONAL SUMMARY

**Senior backend engineer (8+ years) operating as Squad Lead**, specializing in **AWS-deployed distributed systems, Python backend services, and event-driven microservices**. Architected production systems on **AWS (EC2, S3, Lambda, IAM, CloudWatch)** serving 10K+ transactions/day at Almatar; eliminated production crash class via MongoDB re-architecture (40% CPU reduction). Hands-on with **Python (FastAPI, Flask)**, GenAI integration patterns, and prompt engineering. Solo-architected and deployed Zyntra - a real-time social platform on AWS - owning full-stack decisions from schema to production. Strengths in distributed systems, AWS architecture, idempotent service design, and cross-team integrations.

## TECHNICAL SKILLS

**Cloud & AWS:** AWS EC2, S3, Lambda, IAM, CloudWatch, API Gateway, SQS, SNS, RDS, DynamoDB.

**AI / GenAI:** LLM Integration (OpenAI, Anthropic Claude, AWS Bedrock), Prompt Engineering, RAG Architecture, Vector Embeddings, Function Calling, Agentic Workflows

**Languages & Frameworks:** Python (FastAPI, Flask), TypeScript, JavaScript, Node.js, NestJS, Express.js, Flutter (Dart), Angular, React

**Databases & Caching:** MongoDB, MySQL, SQL Server, IBM DB2, Redis, DynamoDB, Drift, Data Modelling, Query Optimization, Indexing

**Messaging & Real-Time:** Apache Kafka, RabbitMQ, AWS SQS/SNS, WebSockets (Socket.IO), SSE, Event-Driven Architecture

**Architecture & Design:** Microservices, HLD/LLD, Distributed Systems, REST API Design, SOLID, Design Patterns (Adapter, Guard, Strategy)

**DevOps & Observability:** Docker, GitLab CI, AWS CloudWatch, Grafana, CI/CD, OAuth 2.0, OIDC, JWT, Logging, Metrics, SLI/SLO

## INDEPENDENT PRODUCT - ARCHITECTURE & ENGINEERING

### Zyntra - Social & Real-Time Messaging Platform

2026 – Present

*Consumer social app - discover/connect, 1:1 chat, posts, relationships. Solo-architected end-to-end; deployed on AWS.*

**Stack:** AWS (EC2, S3, IAM, CloudWatch), NestJS, TypeScript, MongoDB, Redis, Apache Kafka, WebSockets (Socket.IO), Flutter (Drift), Docker

- **AWS-deployed microservices architecture** - split backend into 5 services (auth, user, messaging, matchmaking, notification) deployed on EC2 with Docker, communicating via Kafka event contracts for asynchronous, decoupled workflows. CloudWatch for logs/metrics; IAM for least-privilege service access.
- **Auth, capability framework & multi-level privacy** - three-stage user journey (temporary → registered → verified) with stateless tier claims in JWT; metadata-driven NestJS guard system (@RequiresCapability, @OwnerBypass) for declarative feature access; user-scope and conversation-scope privacy enforced at both query and WebSocket dispatch layers.
- **Secure AWS S3 media pipeline & cache-first Flutter layer** - direct-to-S3 uploads via time-scoped pre-signed URLs offload large payloads from backend; Drift-backed local store renders feed/profile/messages offline-first with background server reconciliation that never blocks the UI.
- **GenAI-ready architecture** - designed pluggable matchmaking/recommendation service with adapter pattern enabling LLM-based ranking (OpenAI / AWS Bedrock) and vector embeddings; prompt templates and model providers abstracted behind unified interface for A/B testing across providers.

## PROFESSIONAL EXPERIENCE

### Senior Software Engineer - Full Stack | Almatar Software Solution Pvt Ltd

Mar 2024 – Present

*Online Travel Agency (OTA) serving 10K+ bookings/day - flight booking, fare search, recommendations, and add-on services.*

- **Own 4 backend microservices on AWS** across booking, fare search, recommendations, and airline integrations - Node.js, NestJS, Python utilities, MongoDB, Redis, Kafka, RabbitMQ, REST APIs; deployed on EC2/ECS with CloudWatch observability. Authored 5+ design documents adopted as squad-level patterns; shipped to production, own on-call.

- **Lead a squad of 5 engineers** on operations enhancements - drive technical direction, HLD/LLD reviews, sprint planning, code-quality standards, and stakeholder alignment with product and platform teams; partner with hiring panels for backend/full-stack roles
- **Led re-architecture of a MongoDB bottleneck** where a single document aggregated all supplier/airline data, driving unbounded growth and service crashes; isolated data into per-supplier/airline documents with backwards-compatible read/write paths; coordinated zero-downtime migration on AWS - **40% lower CPU, 25% lower memory, crash class eliminated**, with downstream AWS compute cost reduction.
- **Co-designed the branded fare recommendation engine** - partnered on architecture for ranking fare families (basic/flex/plus) by traveller intent and business rules; defined feature contracts, ranking signals, and feedback loops - **20% engagement lift, 20% growth in add-on sales**. Exploring LLM-augmented ranking for next iteration.
- **Drove observability, performance & cross-team integrations** - built AWS CloudWatch + Grafana dashboards, alerting against SLO budgets (**20% p95 latency improvement**); led integrations with Ads and User Loyalty platforms (versioned API contracts, event-driven pattern adopted across booking-flow features).
- **Mentor 4-5 junior engineers** on service design and code quality; interviewed 10–15 backend/full-stack candidates; drive squad design discussions and architecture reviews.

#### Senior Software Developer - Full Stack | IBM India Pvt Ltd.

Jun 2022 – Feb 2024

Enterprise telecom automation - International Toll-Free Number (ITN) provisioning on AT&T's Direct Call Provisioning (DCP) system across 60+ countries.

- **Built Node.js + Python microservices** from the ground up with reusable architecture, secure REST API authentication, and cloud-deployable design - **50% automation of provisioning, 90% less manual effort, 70% lower operational cost** across 60+ countries.
- **Improved automation pipeline latency by 20%** via service-level optimizations and streamlined integration contracts with AT&T DCP and carrier back-office systems.
- **Technically led a team of 4 junior engineers** - drove architecture and code reviews, task delegation, mentoring on microservice design, API contracts, and security-by-default; 100% on-time delivery.
- **Owned cloud deployment and container hardening** (Docker, CI/CD); implemented IAM-style security protocols and drove security reviews - **30% reduction in vulnerabilities** aligned with enterprise compliance standards.

#### Software Engineer - Full Stack | Irisidea Tech Solutions Pvt Ltd.

Feb 2021 – May 2022

Client engagement - Boons, US-based hyper-local food ordering & delivery platform (POS integration, rider dispatch, fallback to DoorDash / Uber Eats / Postmates).

- **Designed and built an internal NPM package for multi-provider fallback delivery routing** using the adapter pattern over DoorDash, Uber Eats, and Postmates - normalized divergent auth, contracts, and webhook formats behind a single interface; idempotent dispatch and retry handling to prevent double-shipments under transient failures. Pattern directly reusable for multi-LLM provider abstraction.
- **Built Clover POS integration microservices** (Node.js, Express.js, MongoDB) - order acceptance, terminal creation, lifecycle tracking, payment-capture reconciliation (90%-unit test coverage, 50% fewer post-deploy defects).
- **Sole full-stack engineer on a lean client team** - authored HLD/LLD and design docs; owned solo big-bang migration from AngularJS → Angular 8 (10% faster load, 40+ modules migrated).

#### Project Engineer | Solize India Technologies Pvt Ltd.

Sep 2017 – Jan 2021

- **Built Python desktop tools and automation scripts** (data processing, file pipelines, batch backups) for automotive client Tenneco - 50% effort reduction, 80% less manual maintenance, standardized outputs for downstream design pipelines.
- **Developed an internal full-stack platform** (Node.js + Angular) with REST APIs for engineering workflows; optimized legacy codebase for 30% faster execution on long-running data jobs

## EDUCATION

**Master of Computer Applications (MCA)** - Visvesvaraya Technological University, Bengaluru

2014 – 2017

**B.Sc., Computer Science** - Gulbarga University, Gulbarga

2011 – 2014

## CERTIFICATIONS & AWARDS

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- **Client & Partner Success Award - IBM India (2022)** — led AT&T DCP automation across 60+ countries (90% manual effort reduction, 70% cost savings).
- **Performance Excellence Award - IBM India (2022)** — 50% process automation and 100% on-time delivery on the global voice services program.